

ALBERT LEA, MN, USA

WMO: 726589

Lat: 43.683N

Lon: 93.367W

Elev: 1259

StdP: 14.04

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94968

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.4	-7.8	-18.0	2.2	-10.4	-14.1	2.7	-6.7	26.9	14.0	24.6	13.6	7.7	300	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	89.7	73.6	86.1	72.2	83.6	70.9	77.1	84.6	75.3	82.8	73.6	80.7	9.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	136.6	81.5	72.7	127.2	80.1	71.8	123.0	79.3	41.4	84.5	39.7	82.1	38.1	80.5	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	20.1	18.1	-18.1	93.9	6.3	3.4	-22.6	96.4	-26.3	98.4	-29.8	100.3	-34.4	102.8
			-18.5	80.2	6.1	2.5	-22.8	81.9	-26.4	83.4	-29.8	84.8	-34.2	86.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	16.0	19.7	32.3	46.2	58.2	68.6	72.4	69.4	62.9	49.3	34.9	21.7
	DBStd	22.17	13.19	12.73	12.51	10.37	8.94	6.68	5.48	5.43	8.39	9.62	11.38	11.89
	HDD50	4161	1054	848	564	188	25	0	0	0	7	131	467	877
	HDD65	7559	1519	1268	1015	567	248	39	7	19	138	495	902	1342
	CDD50	2741	0	0	15	75	281	557	693	603	394	109	14	0
	CDD65	663	0	0	0	4	39	146	235	157	74	8	0	0
	CDH74	5547	0	0	2	62	389	1245	2016	1135	614	83	1	0
	CDH80	1563	0	0	0	16	120	378	614	263	155	17	0	0
Wind	WSAvg	8.0	8.8	8.9	9.3	9.9	8.8	7.1	5.9	5.4	6.6	8.1	8.8	8.5
Precipitation	PrecAvg	31.6	0.7	0.7	1.8	3.1	4.1	4.5	4.4	4.0	3.3	2.3	1.6	0.9
	PrecMax	50.7	2.3	2.8	4.1	7.8	8.2	10.1	10.4	10.5	9.1	6.9	5.5	2.7
	PrecMin	17.0	0.0	0.0	0.2	0.6	1.2	1.4	1.0	0.7	0.5	0.2	0.0	0.0
	PrecStd	6.9	0.6	0.6	1.1	1.7	1.9	2.1	2.4	2.1	2.2	1.5	1.3	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	53.8	71.7	81.8	89.8	92.5	92.8	89.8	89.6	81.9	69.7	53.9
		MCWB	39.8	47.6	58.6	61.3	67.8	72.8	76.6	75.8	71.4	65.2	56.1	50.0
	2%	DB	39.2	45.3	62.8	73.1	82.4	88.1	89.7	85.6	84.0	74.9	62.5	45.2
		MCWB	35.5	41.6	54.5	56.4	63.7	71.5	75.7	74.0	70.6	61.1	53.3	41.3
	5%	DB	36.3	39.4	55.3	68.0	78.7	84.0	86.1	82.3	81.2	70.1	55.3	39.4
		MCWB	34.2	36.4	48.8	53.9	62.0	68.8	74.5	72.2	68.0	59.3	48.4	36.4
	10%	DB	33.9	36.5	48.3	63.3	73.1	81.3	82.4	80.9	77.2	64.0	51.6	36.7
		MCWB	32.5	34.0	44.0	51.6	59.4	67.7	72.0	70.9	65.8	55.4	45.9	34.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.6	47.3	61.4	63.6	72.3	77.0	81.5	78.2	75.4	70.5	58.6	50.1
		MCDB	45.9	52.0	68.0	74.7	82.0	85.7	86.3	85.4	83.2	77.8	65.7	54.2
	2%	WB	36.6	41.3	55.5	59.5	67.7	74.6	78.0	76.1	73.0	64.8	54.2	41.3
		MCDB	39.1	45.1	60.8	69.5	77.6	83.1	85.3	82.7	80.4	70.5	60.1	44.7
	5%	WB	34.3	37.2	49.7	56.1	65.1	72.3	76.1	74.3	70.6	60.5	49.8	37.2
		MCDB	36.0	39.8	56.4	65.6	73.6	80.9	83.7	80.6	77.7	67.4	56.3	39.5
	10%	WB	32.0	34.2	44.4	53.0	62.5	70.0	74.3	72.3	68.2	56.7	45.6	34.7
		MCDB	33.7	36.4	48.6	61.2	70.3	78.5	81.6	78.5	75.1	63.5	49.8	36.7
Mean Daily Temperature Range	5% DB	MDBR	15.2	15.5	15.8	19.2	19.6	18.3	17.9	18.0	20.4	18.9	15.5	13.6
		MCDBR	17.0	16.9	24.1	28.6	25.2	21.9	19.5	19.4	23.0	26.0	22.6	16.7
		MCWBR	14.3	13.5	15.9	15.0	11.2	9.8	9.9	10.3	11.2	13.7	14.5	13.5
	5% WB	MCDBR	15.6	15.9	22.7	24.1	20.3	18.6	17.5	17.2	19.6	20.6	20.9	14.8
	MCWBR	13.6	13.0	15.6	14.8	11.6	11.0	10.3	10.5	11.1	13.7	14.3	12.5	
Clear-Sky Solar Irradiance	taub		0.266	0.288	0.311	0.366	0.393	0.415	0.417	0.404	0.374	0.326	0.289	0.266
	taud		2.429	2.364	2.427	2.318	2.302	2.281	2.270	2.300	2.355	2.485	2.509	2.453
	Ebn at Noon		278	289	296	285	279	272	270	270	270	272	268	268
	Edh at Noon		23	30	32	39	40	42	42	39	34	26	22	21
All-Sky Solar Radiation	RadAvg		566	902	1190	1469	1724	1926	2019	1699	1359	886	582	425
	RadStd		61	93	118	155	132	149	89	120	95	117	50	30

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page